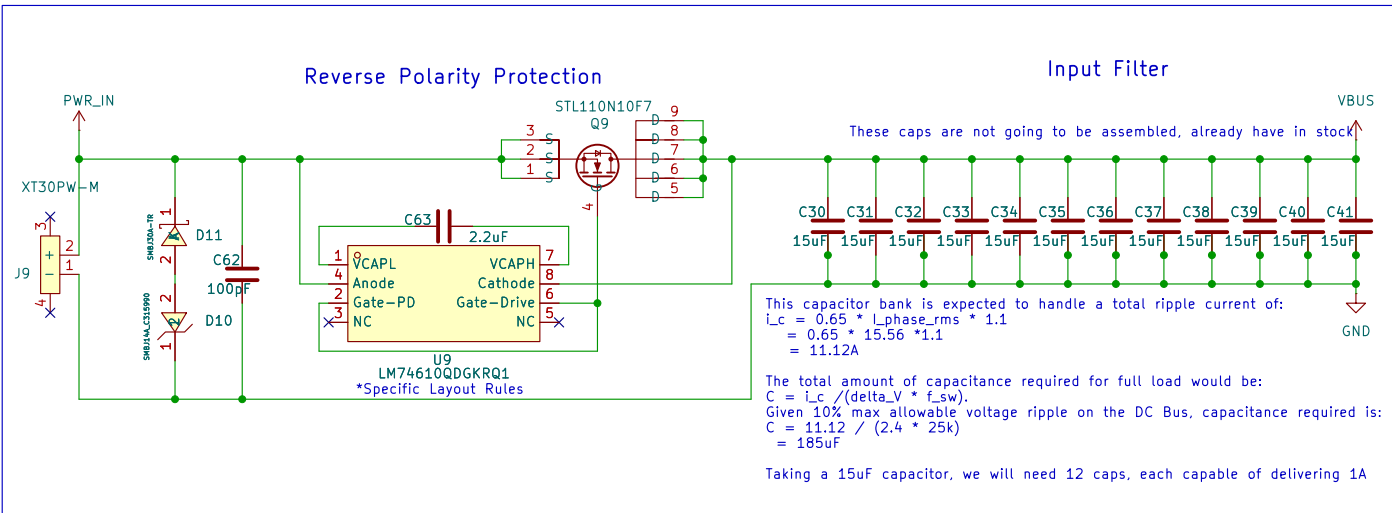
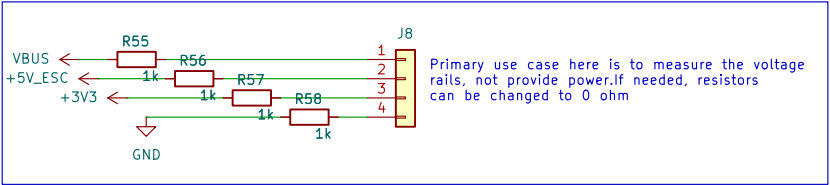




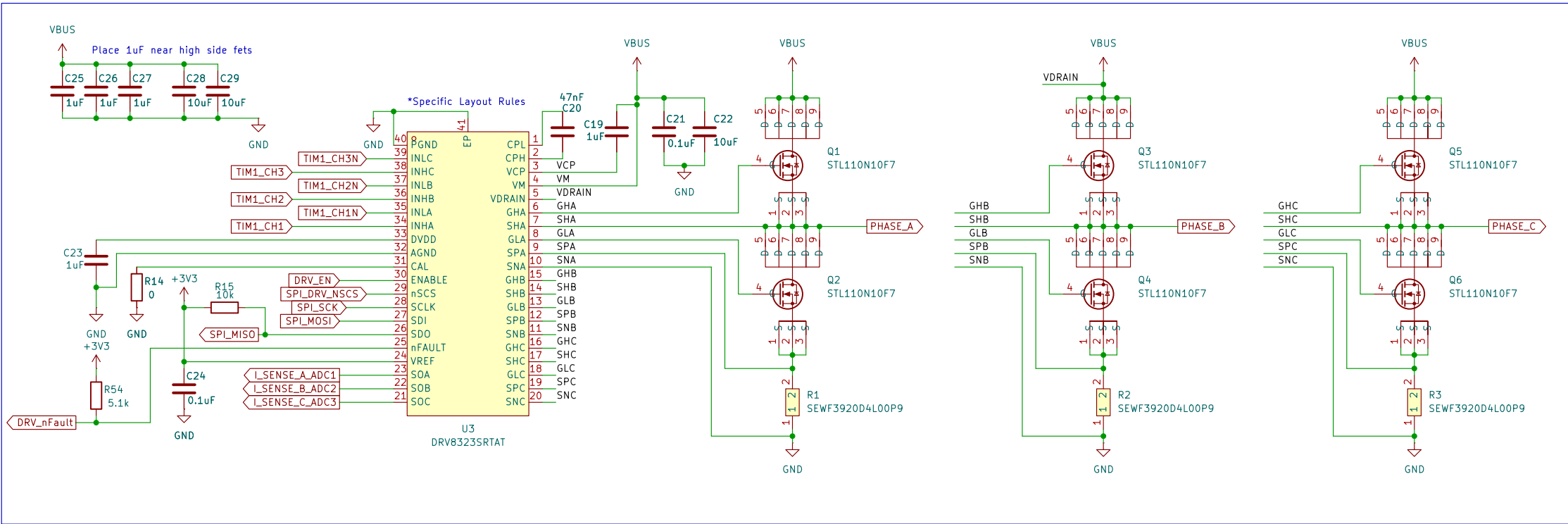
INPUT POWER



EXPOSED POWER RAILS



INVERTER



Current Sense Discussion

For this esc setup, the motor driver ensures that $V_{REF}/2$ is the middle reference point for the current sensing, where here V_{REF} is 3V3. From here, a gain of 10 is selected, in order to provide the mcu with an idle voltage of 1.65V when there is no current flowing through a phase. This means, that under peak load (22A), we can expect a max voltage (assuming 4m shunt) of:

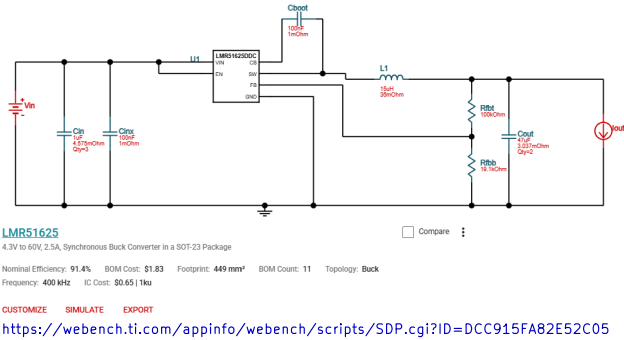
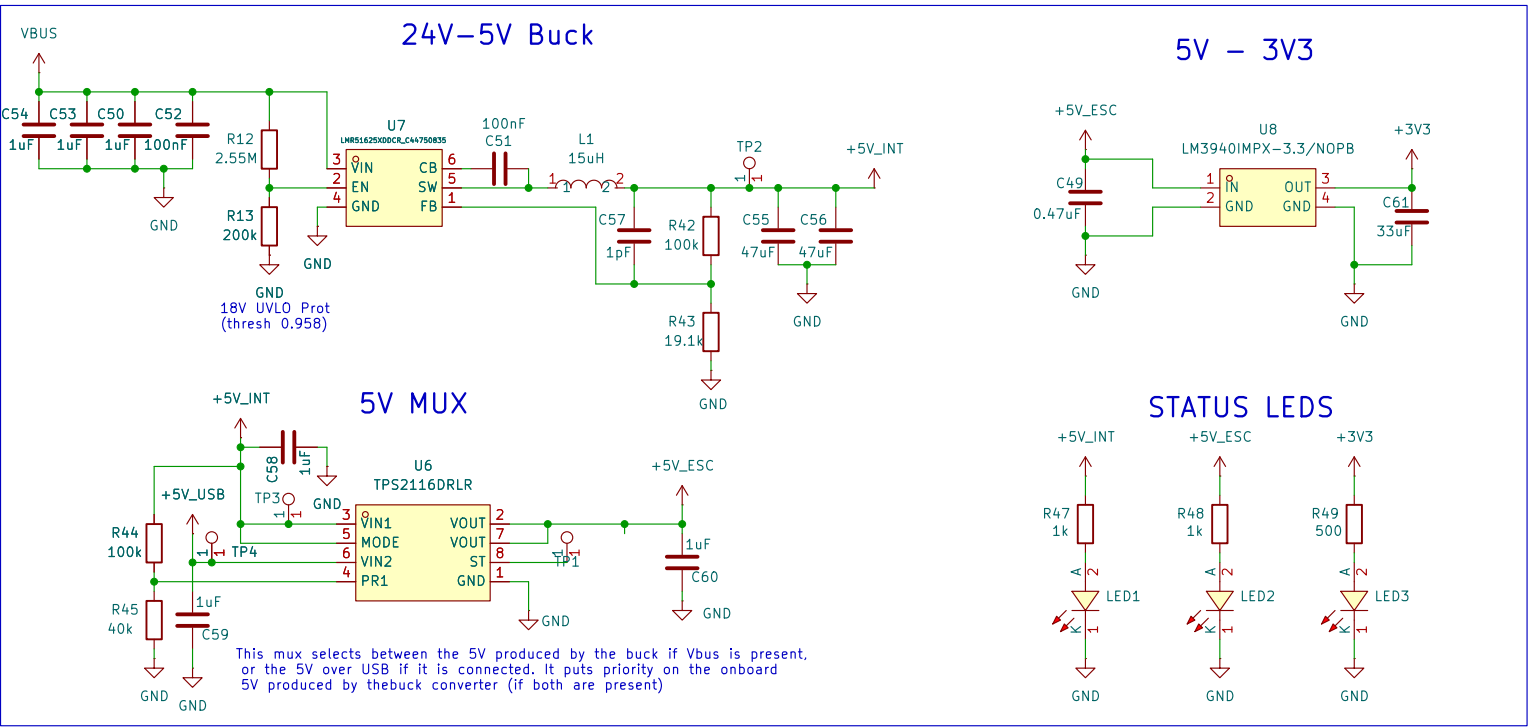
$$\begin{aligned} V_{out} &= 1.65 + (\text{Gain} \cdot I \cdot R_{shunt}) \\ &= 1.65 + (10 \cdot 22 \cdot 0.004) \\ &= 2.53V \end{aligned}$$

This is below the maximum of 3V3, which is ideal. This is because we need some headroom to detect if ever there is an overcurrent event in one of the phases. If this was too close, it would either have false detections, or be hard to detect such an event.

We can also calculate the maximum power dissipated through this shunt as follows

$$\begin{aligned} P &= I^2 \cdot R_{shunt} \\ &= (22)^2 \cdot 0.04 \\ &= 1.94W \end{aligned}$$

DC-DC CONVERTERS + 5V MUX



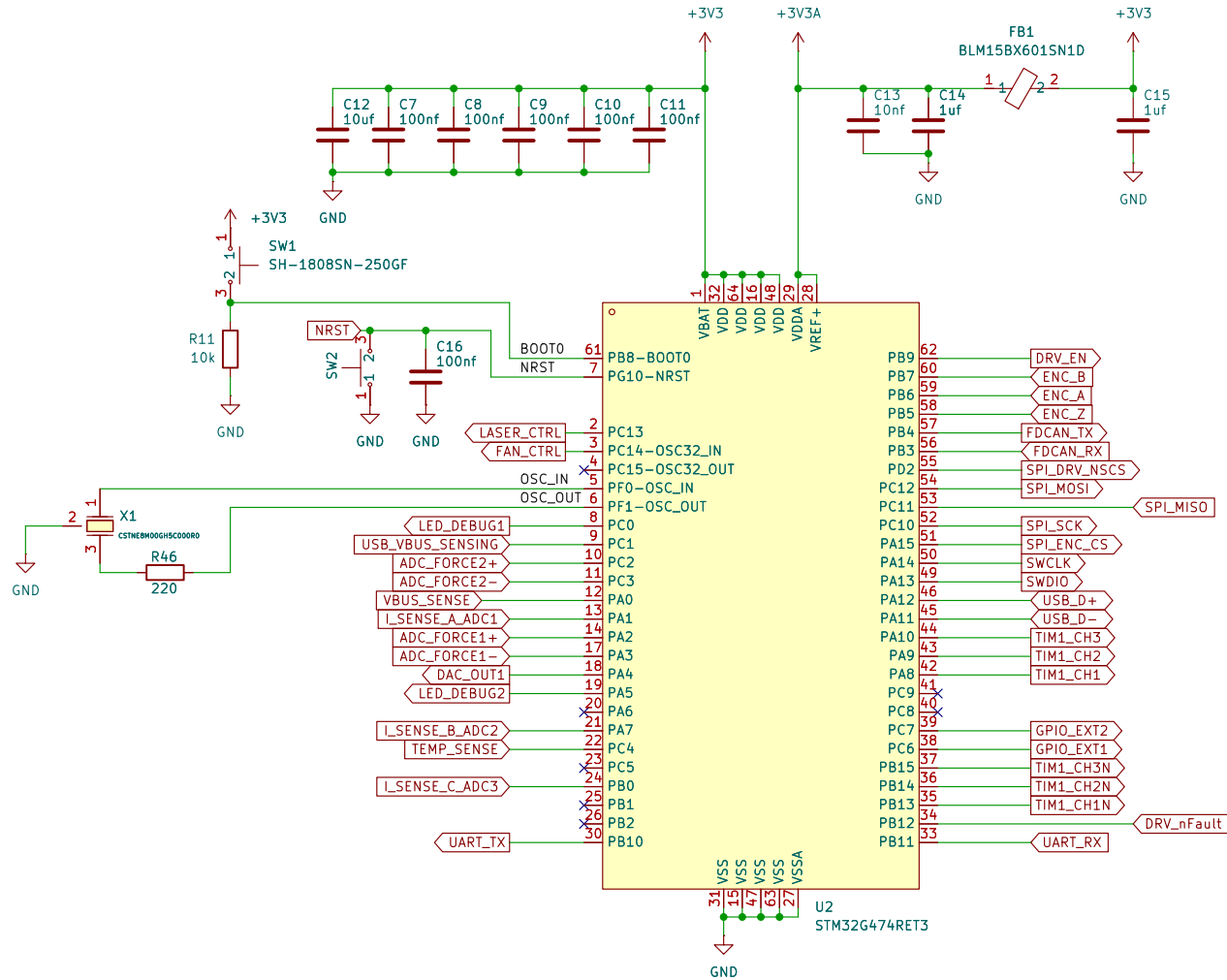
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KiCad E.D.A. 10.0.1

Rev: V1
Id: 2/4

MCU



Sheet: /mcu/
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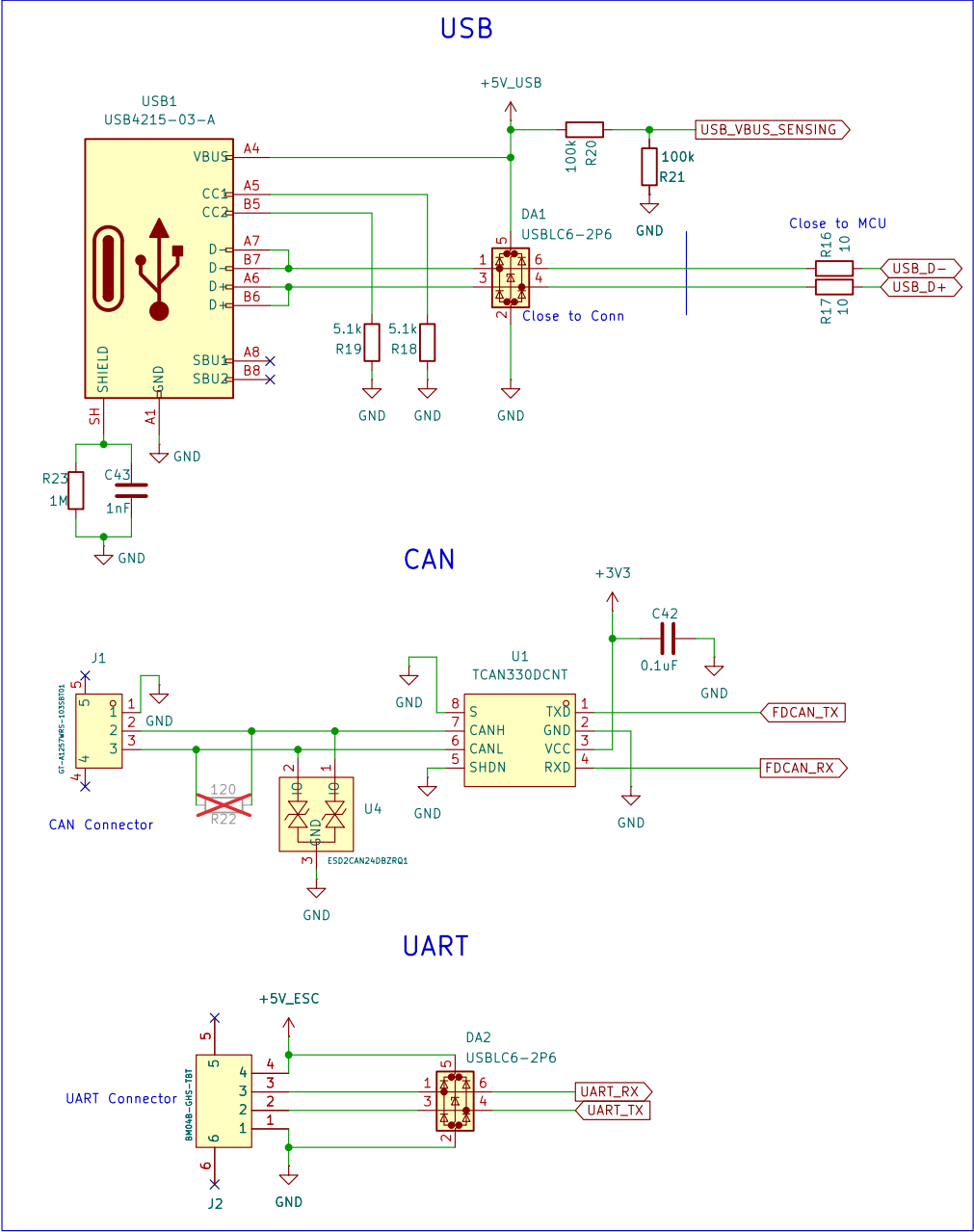
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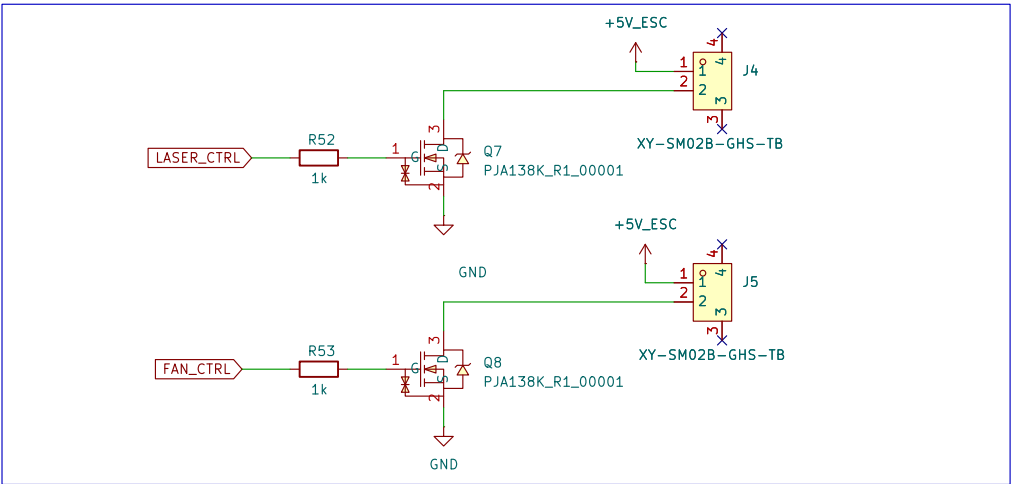
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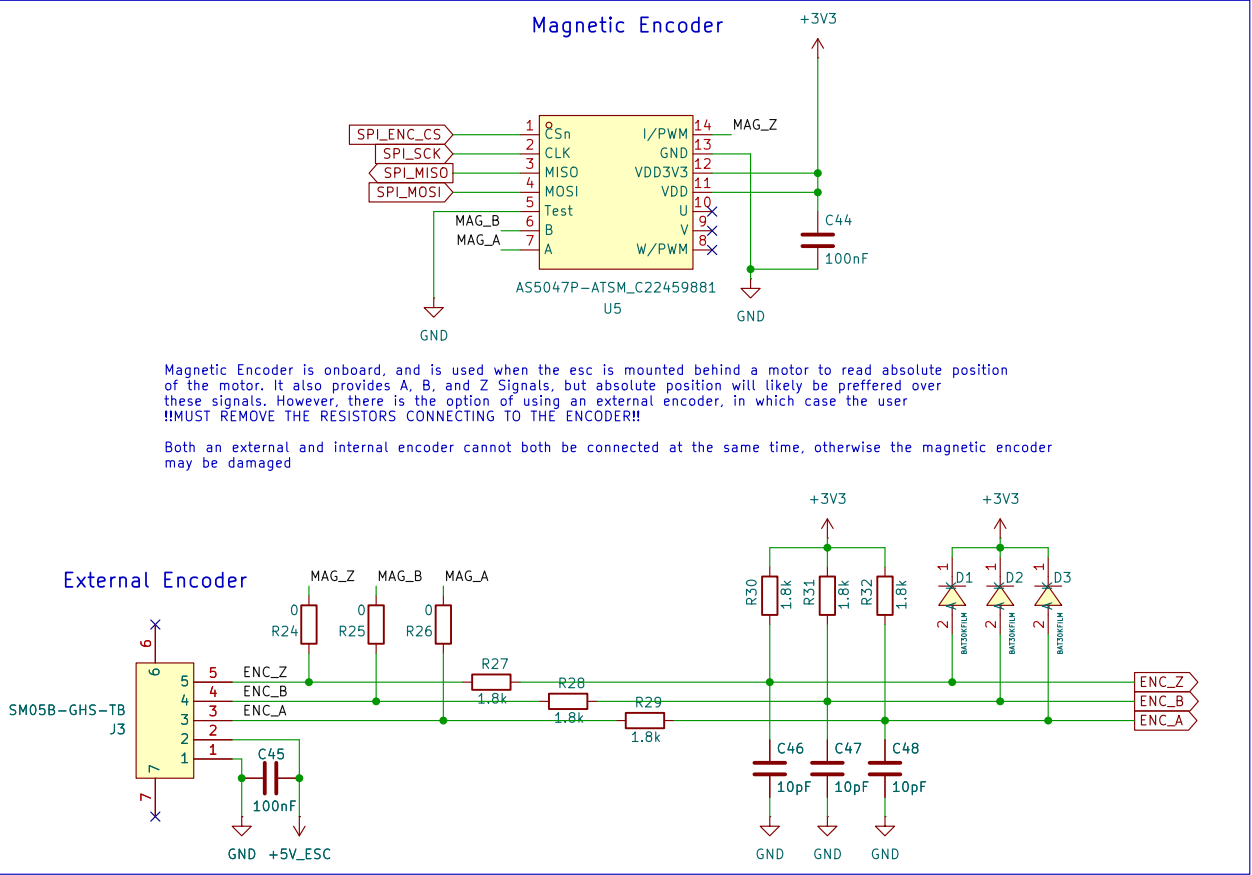
COMMUNICATION



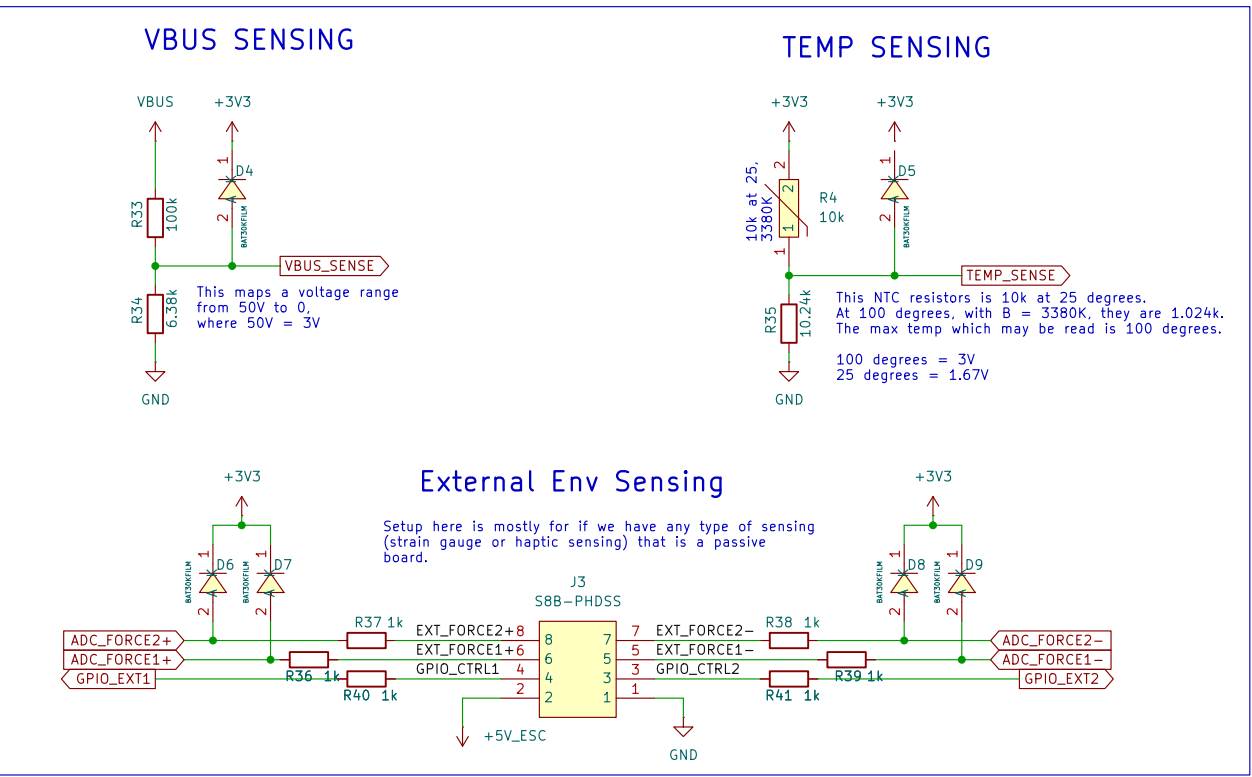
FAN AND LAZER CONTROL



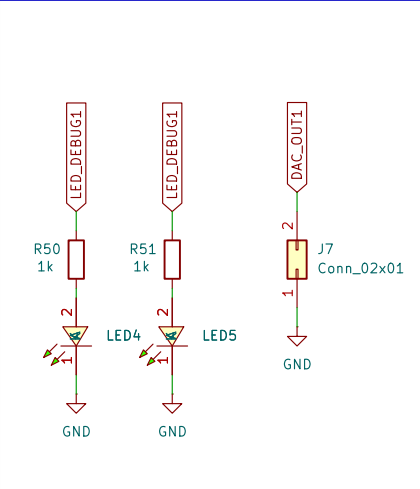
ENCODERS



MCU SENSING



DEBUG



FLASHING

